Higher National Diploma in Engineering (Electrical)
4th Year, First Semester Examination – 2018
EE 3214 – Construction and Protection of Electrical systems

Instructions for Candidates:
answer **only five** questions.
each question carry equal marks

No. of questions : 06
No. of pages : 04
Time : **03 hours**

1.

i. Write three advantages of per unit system for power system analysis

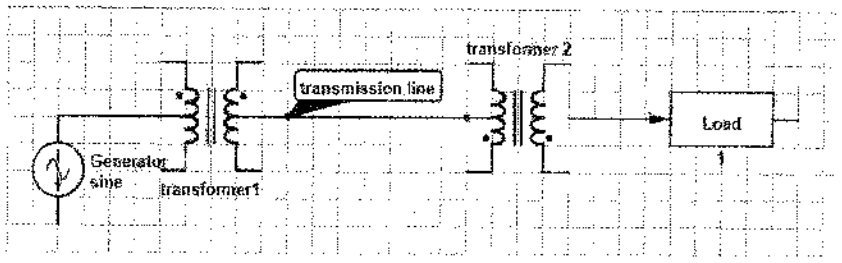
3 Marks

ii. Derive expression for the Base current and Base impedance by using Base KVA and Base Voltage. 4 Marks

The generator is 11kV, 30 MVA with per unit reactance 0.15.
The generator supplies power to a load through a transmission line having transformer at both ends. Transformers are 0.3 p.u reactance, 10 MVA, 11/110 kV and 0.2 p.u reactance, 10 MVA, 110/11 kV respectively.
Load has 0.1 per unit reactance at 10MVA, 11 kV.
Series reactance of transmission line is 60 ohms.

a. Draw the reactance diagram of the system shown in figure 1. For selecting 30MVA for the complete system.
b. Per unit current go through Electrical System.
c. Actual current go through the transmission line

13 Marks



(Total Marks 20)

2.

i. Write a expression to Fault MVA and Fault current in three phase system for symmetrical faults 4 Marks

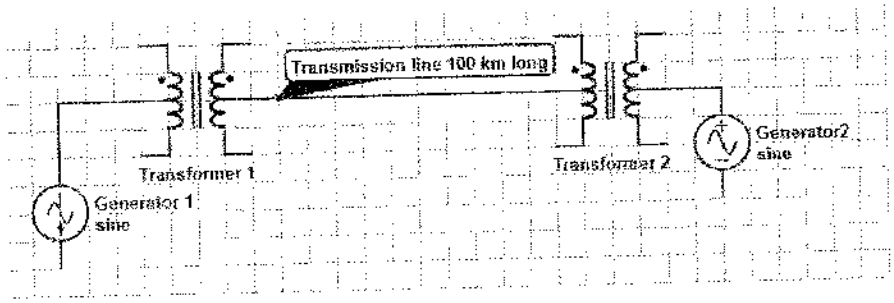
ii. Two generators are rated 11kV, 6 MVA and having 10% reactance are interconnected by 100 km long transmission line. The reactance of transmission line is 0.2 ohms per km. Transformers near the generators are rated 3 MVA, 11kV/66kv and have 5% reactance each.(As shown in fig. 2.)

A three phase symmetric fault occur at a distance from 30km from one end of the transmission line when the system is on load but at rated voltage.

Calculate (Selecting 6 MVA as common base MVA)

a. Fault MVA

b. Fault Current 16 Marks



(Total Marks 20)

3.

i. Write three types of unsymmetrical faults in power system 3 Marks

ii. Define operator 'a'(or λ) used in unsymmetrical fault calculation 2 Marks

iii. Show that $1 + a + a^2 = 0$ 3 Marks

iv. Write expression to represent three current components in lines(R,Y.and B) by using positive, negative and zero components of red line.

v. A three phase 4 wire system supplies loads which uniquely distributed in the three phases. Positive sequence current of R line is $300 < 0^0$ A and negative phase sequence component is $130 < 70^0$and the total current following back to the supply in the neutral conductor is$275 < 238^0$. Calculate the current in the three lines. 12 Marks

(Total Marks 20)

4.

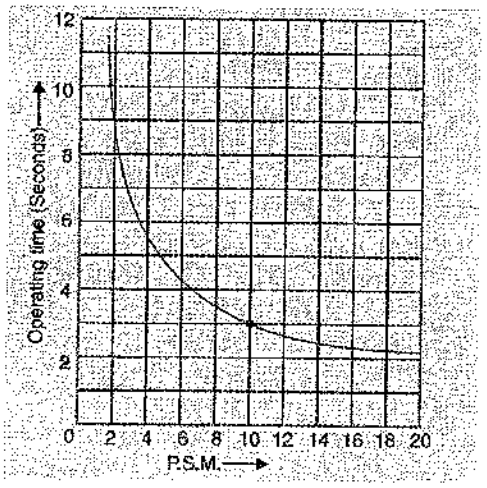
i. What is the operating principle of a relay ? 2 Marks

ii. Write 3 qualities of a relay 3 Marks

iii. Define Pick up current, Current setting, Plug setting multiplier (PSM), Time setting multiplier (TSM), Time/P.S.M Curve 5 marks

iv. The relay is used to protect a feeder of the three phase power system through a 100/5A current transformer. Calculate the time of operation of the relay when feeder fault current is 2100A. The relay has current Setting of 150% and Time Setting of 0.4. Nominal current of the relay is 5 A. The relay has the following characteristics.

10 Marks



(Total Marks 20)

5.

i. What are the two common faults of a transformer? 2 Marks

ii. What are the three faults on generator? 3 Marks

iii. Explain the principle of Merz-price circulating current system of protection used for transformers with suitable sketch.

5 Marks

iv. A 3-Phase, 11/33 kV, delta-star connected power transformer is protected by differential protection (Merz-price circulating current system). The current transformer on the low Voltage side has a current ratio of 500/5 A.

a. How Current transformers should be connected (Draw the arrangement of power transformer and current transformers).

b. What must be the ratio of CT's on the HV side 10 Marks

(Total Marks 20)

6. Write short note
   i. Construction and operation of Buchholz relay
   ii. Distance protection principle
   iii. Faults on Motor and motor protection system
   iv. Operation of Air blast circuit breaker
   v. HRC Fuse

5Marks x 4

(Total Marks 20)